

Space, Rank & Basis

Matrices, Column space, Null space

A matrix is like a linear function from vectors to vectors.

- Ax is the linear combination of matrix A with vector of weights x .

$$\begin{matrix} \begin{bmatrix} 2 & 3 & 1 \\ 1 & 1 & 5 \end{bmatrix} & \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} & = & \begin{bmatrix} 2x-1+3x0+1x2 \\ 1x-1+1x0+5x2 \end{bmatrix} & = & \begin{bmatrix} 0 \\ 9 \end{bmatrix} \\ 2 \times 3 & 3 \times 1 & & \text{using dot product} & & \end{matrix}$$

$$= \begin{bmatrix} 2 \\ 1 \end{bmatrix} \times (-1) + \begin{bmatrix} 3 \\ 1 \end{bmatrix} \times 0 + \begin{bmatrix} 1 \\ 5 \end{bmatrix} \times 2 = \begin{bmatrix} 0 \\ 9 \end{bmatrix}$$

using column view

- The column space $C(A)$ is where b lives in $Ax=b$
For an $m \times n$ matrix, $C(A)$ is a subspace of $\mathbb{R}^m$ of dim. $\text{rank}(A)$.
- The nullspace $N(A)$ is where x solves $Ax=0$
For an $m \times n$ matrix, $N(A)$ is a subspace of $\mathbb{R}^n$ of dim $n - \text{rank}(A)$.

Rank

The rank is equal to the number of pivot columns in a matrix.
Say A is equal to:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 4 & 7 \end{bmatrix} \xrightarrow{\text{elimination}} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -2 & 1 \end{bmatrix} \xrightarrow{\begin{matrix} -2R_1 \\ -3R_1 \end{matrix}} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -2 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} \text{Two pivots} \\ \Rightarrow \text{rank}(A) \text{ is } 2 \end{matrix}$$

Basis

A basis for a space is a set of vectors linearly independent and that spans the space.

A set of vectors is independent if its nullspace = $\{\vec{0}\}$

Ambient space vs. Dimension

" $C(A)$ is an r -dimensional subspace of $\mathbb{R}^m$ "

" $N(A)$ is an $(n-r)$ -dimensional subspace of $\mathbb{R}^n$ "

The ambient space is how many entries a vector has,
the dimension is how many independent directions it has.